



Regnant batteries

At present, the demand-supply gap is met through imports. However, in the times to come, India will see a big expansion in its manufacturing capacity.

At present, SMF batteries are 20 per cent costlier than conventional lead-acid batteries. However, through mass production, costs can be brought down. "SMF batteries require a different type of separator and plate content. Also, there are more manufacturing steps. At present, separator is imported from countries like USA and Europe as the technology for separators is not developed well. Su-Kam is talking to people in India to develop separators in order to bring down the costs," informs Sachdev.

Advantages over conventional batteries

SMF batteries are far superior to the conventional batteries. Conventional

batteries require regular maintenance, especially topping up of electrolyte and water at regular intervals.

VRLA batteries, on the other hand, require no topping up. "These don't contain acid in the liquid form but in the form of gel, so there is no spillage of acid and you don't need to top up the battery with acid periodically. Their plates are made of lead-calcium, which offers more life and efficiency due to low self-discharge," says Kapil Sood, managing director, Amptek.

SMF batteries are also environment-friendly, require very less space and can be used in any position. "These have very low permeability, low-resistance current path and fast recharge capability," adds Rajesh Gupta,

director, Okaya Power.

In sealed batteries, due to chemical reactions inside the battery, gases are formed. Sometimes due to overcharging, more gases are formed, which need to be released from the battery container. This is done by using valves that release gases under certain pressure.

However, unlike conventional batteries, VRLA batteries cannot be used in extreme conditions, since water loss cannot be compensated by topping up. For such environments, special construction is required and batteries need to be maintained in a controlled environment.

Availability of mains power varies widely across India. Duration of daily cumulative power cut varies from 30 minutes to 12 hours. Any one particular type of battery design cannot operate optimally in such a wide usage condition. A lead-acid battery is a de-

There is shortage of supply, as the production base in India is quite small

—Amitansu Satpathi, director,
Best Power Equipment



3rd

**Largest UPS
Manufacturer of India**

climbing from the 10th to the 3rd position
in just 5 years*

► Full Range of UPS Systems

► Wide Customer Base

► All India Sales & Service Network

Uniline Energy Systems (Pvt) Ltd., Uniline House, Ramesh Market, East of Kailash, New Delhi - 110 065. Tel. : - 011 - 4666 1111, 2646 9031, 2646 9108
Chandigarh : 92167 63793 Chennai : 99400 49677, 98841 42866, Dehradun : 98972 72449 Jaipur : 98290 60772 Kanpur : 94151 32093, 93368 55700 Kolkata : 94330 33855, 98833 59262

D SERIES

High Power IGBT UPS Upto 1800 KVA

GIGALINE

IGBT UPS Upto 200 KVA

MAXILINE

IGBT UPS Upto 100 KVA

GREENLINE

Online UPS Upto 6 KVA

SMARTLINE

Offline UPS Upto 3 KVA

POWERLINE

CVT - SERVO Upto 1000 KVA